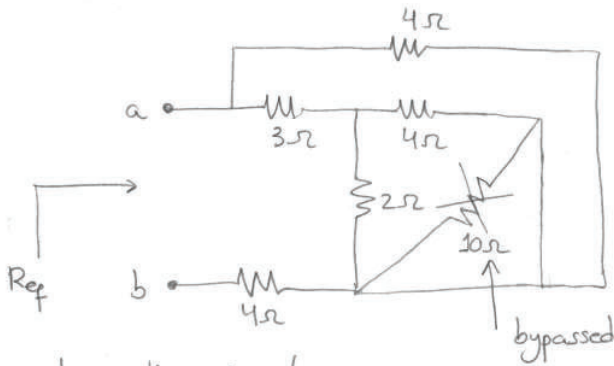
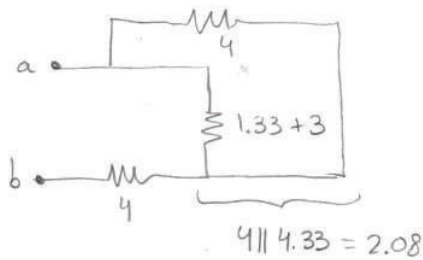
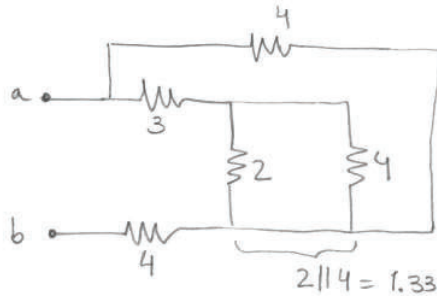


QUESTION 1

a)



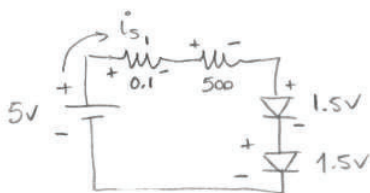
If we re-draw the circuit:



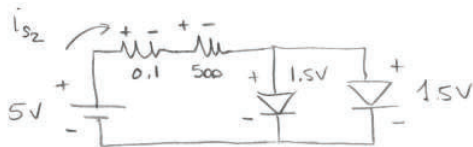
$$R_{eq} = 4 + (4 \parallel 4.33) = 6.08 \Omega$$

b)

i)



$$i_{s1} = \frac{5-3}{500.1} \approx 4 \text{ mA}$$



$$i_{s2} = \frac{5-1.5}{500.1} \approx 7 \text{ mA}$$

ii)

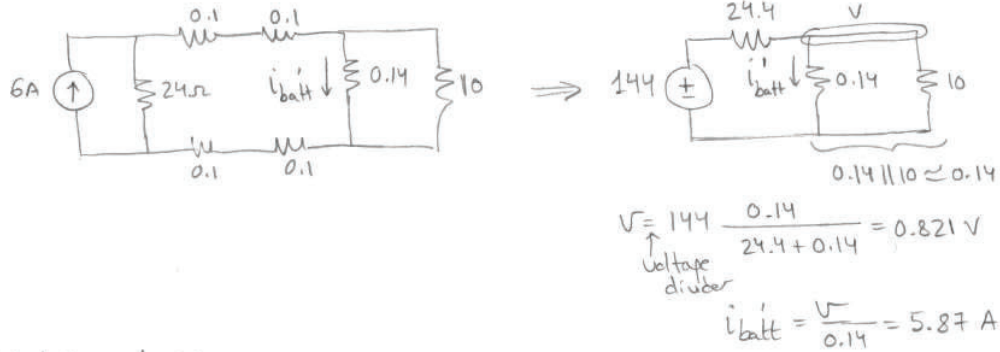
Configuration b)

since it is drawing more current from the battery, and for the given charge of the battery, it will take shorter to discharge (think about amp-hours)

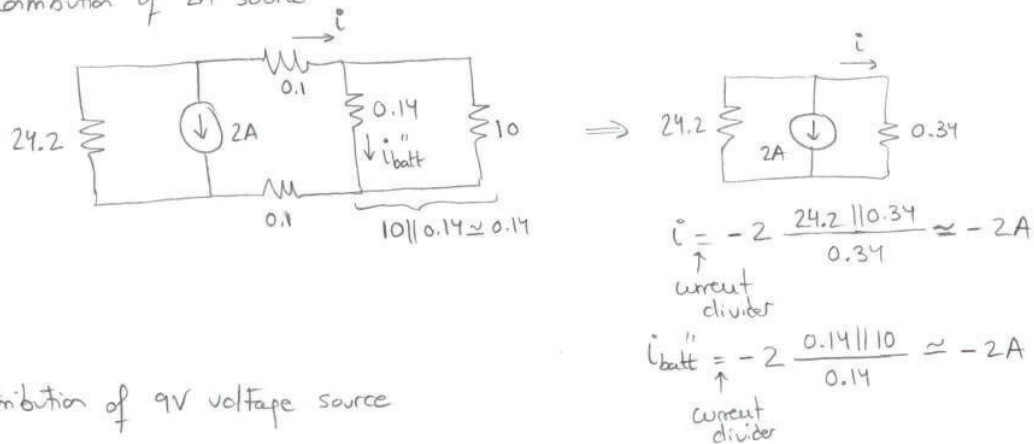
QUESTION 2

2)

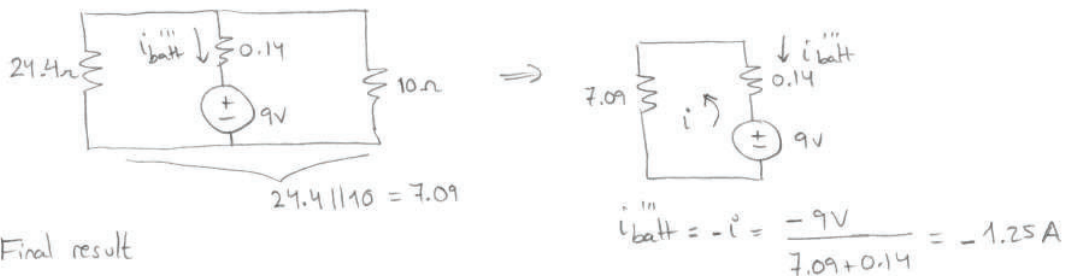
* Contribution of 6A current source



* Contribution of 2A source



* Contribution of 9V voltage source



* Final result

$$i_{batt} = i'_{batt} + i''_{batt} + i'''_{batt} = 5.87 - 2 - 1.25 = 2.62 \text{ A}$$

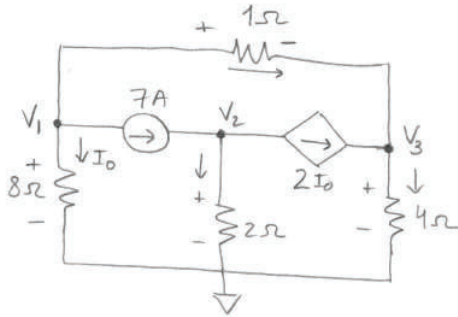
b) $P_{V_{batt}} = V_{batt} \times i_{batt} = 9 \times 2.62 = 23.58 \text{ W}$

$P_{R_{batt}} = i_{batt}^2 \times R_{batt} = 2.62^2 \times 0.14 = 0.96 \text{ W}$

$\Rightarrow P_{batt} = P_{Vs} + P_R = 24.54 \text{ W}$

QUESTION 3

a)



KCL @ node 1:

$$-\frac{V_1}{8} - 7 - \frac{V_1 - V_3}{1} = 0 \Rightarrow -\left(\frac{1}{8} + 1\right)V_1 + V_3 = 7 \Rightarrow -\frac{9}{8}V_1 + V_3 = 7$$

KCL @ node 2:

$$7 = \frac{V_2}{2} + 2I_0 \quad \text{where} \quad I_0 = \frac{V_1}{8} \Rightarrow \frac{V_1}{4} + \frac{V_2}{2} = 7$$

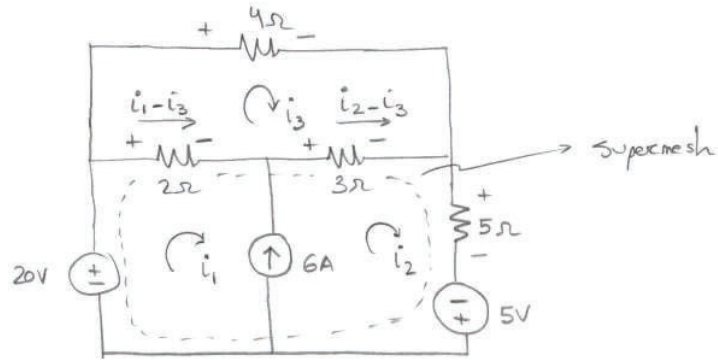
KCL @ node 3:

$$2I_0 + \frac{V_1 - V_3}{1} = \frac{V_3}{4} \quad \text{where} \quad I_0 = \frac{V_1}{8} \Rightarrow \left(\frac{1}{4} + 1\right)V_1 - \left(1 + \frac{1}{4}\right)V_3 = 0; \quad \frac{5}{4}V_1 - \frac{5}{4}V_3 = 0$$

$$\text{Nodal eq. } \begin{cases} -\frac{9}{8}V_1 + V_3 = 7 \\ \frac{1}{4}V_1 + \frac{1}{2}V_2 = 7 \\ \frac{5}{4}V_1 - \frac{5}{4}V_3 = 0 \end{cases}$$

QUESTION 3

b)



* KVL @ supermesh:

$$-20 + 2(i_1 - i_3) + 3(i_2 - i_3) + 5i_2 - 5 = 0$$

$$\underline{\underline{2i_1 + 8i_2 - 5i_3 = 25}}$$

* KVL @ mesh 3:

$$4i_3 - 3(i_2 - i_3) - 2(i_1 - i_3) = 0$$

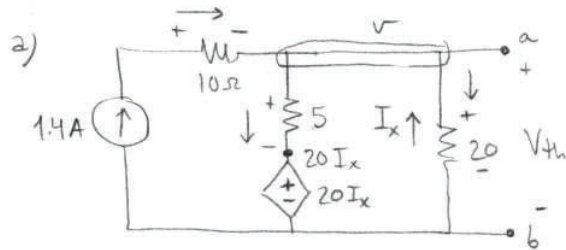
$$\underline{\underline{-2i_1 - 3i_2 + 9i_3 = 0}}$$

* Constraint equation:

$$\underline{\underline{i_2 - i_1 = 6}}$$

$$\text{Mesh eq. } \begin{cases} 2i_1 + 8i_2 - 5i_3 = 25 \\ -2i_1 - 3i_2 + 9i_3 = 0 \\ i_2 - i_1 = 6 \end{cases}$$

QUESTION 4

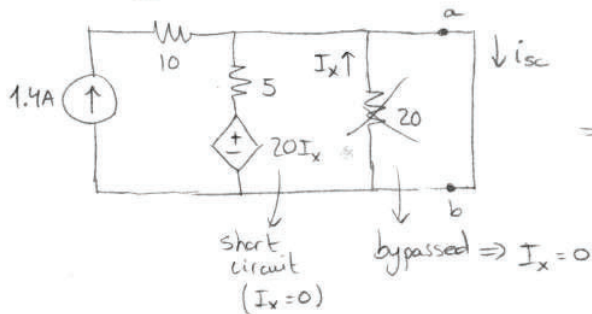


Nodal:

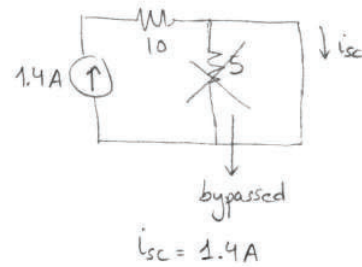
$$1.4 + I_x = \frac{V - 20I_x}{5} \quad \text{where } I_x = -\frac{V}{20}$$

$$\frac{V}{5} + \frac{4}{20}V + \frac{V}{20} = 1.4; \quad \left(\frac{1}{5} + \frac{4}{20} + \frac{1}{20}\right)V = 1.4; \quad \frac{9V}{20} = 1.4 \Rightarrow V = 3.111V = V_{th}$$

b) $R_{th} = V_{th}/i_{sc}$



$\Rightarrow$

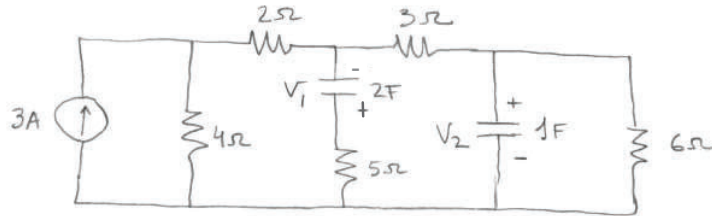


$$R_{th} = V_{th}/i_{sc} = \frac{3.111}{1.4} = 2.22\Omega$$

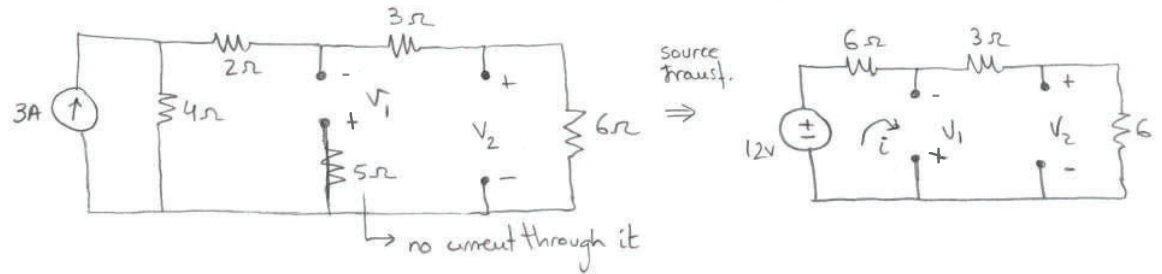
c) $P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{3.111^2}{4 \times 2.22} = 1.089W$

QUESTION 5

2)



In DC steady state, capacitors behave as open circuits



$$\text{KVL @ mesh: } -12 + 6i + 3i + 6i = 0$$

$$-12 + 15i = 0$$

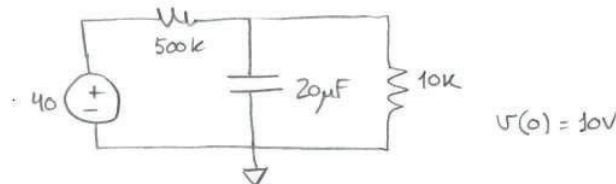
$$i = 12/15 = 0.8 \text{ A}$$

$$\text{KVL in left mesh to calculate } V_1: -12 + 6i - V_1 = 0 \Rightarrow \boxed{V_1 = -12 + 6i = -12 + 6 \times 0.8 = -7.2 \text{ V}}$$

$$\text{KVL in right mesh to calculate } V_2: -V_2 + 6i = 0 \Rightarrow \boxed{V_2 = 6i = 6 \times 0.8 = 4.8 \text{ V}}$$

b)

At $t=0$ we have

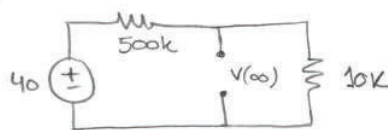


This is a step response:

$$V_c(t) = V(\infty) + (V(0) - V(\infty))e^{-t/\tau}$$

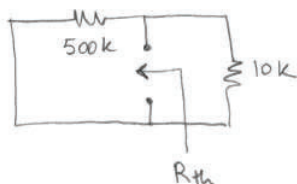
- $V(0) = 10 \text{ V}$

- $V(\infty):$



$$V_{\infty} = V_{10k} = 40 \frac{10k}{500k + 10k} = \frac{400}{510} = 0.784 \text{ V}$$

- $\tau = R_{th}C$



$$R_{th} = 500k \parallel 10k = 9.8k$$

$$\tau = 9.8k \times 20\mu F = 196 \times 10^{-3} \text{ s} = 196 \text{ ms} = 0.196 \text{ s}$$

Substitute:

$$V_c(t) = 0.784 + (10 - 0.784)e^{-t/0.196} \text{ V}$$

$$= 0.784 + 9.216e^{-t/0.196} \text{ V}$$

The lamp will stop conducting when the voltage drops to 2V, so

$$0.784 + 9.216 e^{-t/0.196} = 2$$

$$9.216 e^{-t/0.196} = 1.216$$

$$+t/0.196 = +\ln\left(\frac{1.216}{9.216}\right) = +2.025$$

$$\boxed{T = 2.025 \times 0.196 = 0.397 \approx 0.4 \text{ s}}$$